

Yijing Wu

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SKILLS

Programming Languages: Python, JavaScript, TypeScript, Bash, SQL, Java, HTML, CSS, Swift, C/C++
Frameworks: React, React Native, Node.js, Django, Express, Next.js, Vue.js, Redux, GraphQL, LangChain
Cloud and DevOps: AWS (Lambda, ECR, DynamoDB, EC2, VPC, IAM), Azure, Terraform, Jenkins, Docker, Linux
Others: PostgreSQL, MySQL, MongoDB, Git, Open AI, CocoaPods, RESTful API, PyTest, Unittest, ts-Jest, Jupyter Notebook

EXPERIENCE

Autodesk San Francisco, CA (Remote)
Cloud DevOps Engineer Intern May 2023 – Present

- Built a serverless Slack bot in Python, automating issue resolution by **Generative AI** techniques to support Azure migration
- Integrated ServiceNow and JIRA with Slack APIs to streamline ticket management, resulting in a significant reduction in support response times
- Applied **Azure OpenAI** and **LangChain**, utilizing **prompt** engineering, **embeddings**, **NLP** and **LLM Agents** to achieve retrieval augmented generation (RAG), data analysis and interactive conversations
- Established a robust **CI/CD** pipeline using **Terraform** and **Jenkins**, following Infrastructure as Code (**IaC**) principles to enable end-to-end cloud deployment automation, thereby enhancing efficiency and maintaining consistency
- Implemented **serverless** architecture in **AWS**, utilizing services such as **Lambda**, **ECR**, **S3** Bucket, **DynamoDB**, **SageMaker**, **VPC**, **API Gateway**, **IAM**, and **CloudWatch** to optimize the backend infrastructure for seamless operations

Upldo San Francisco, CA (Remote)
Full-Stack Software Engineer Intern June 2022 – December 2022

- Contributed as a full-stack engineer in an agile team to build, enhance and maintain cross-platform web & mobile application
- Collaborated with product and design team, conducting front-end development in **React** and **React Native** along with **Redux** through **TypeScript** to enhance UI/UX, delivered graph data visualization, animations, and kudos interface
- Worked on **Django** backend (Python), applied **Celery** for asynchronous task management, optimized **GraphQL** to minimize queries to **PostgreSQL** database, and built **Unit Testing** and **Integration Testing** to ensure application robustness
- Implemented a video call notification feature using **iOS** framework **CallKit** through **Objective-C** in **Xcode** and **Android** **ConnectionService** through **Java** in **Android**, leading to a significant enhancement in match pick-up rate
- Performed camera and microphone selection and checking features through **Agora** to ensure **WebRTC** video calling quality

Huawei Technologies Shanghai, China
Algorithm Engineer Intern July 2020 – September 2020

- Applied **computer vision** techniques to calibrate and correct camera sensor distortion in self-driving vehicle video streams
- Implemented cylindrical projection and single-point mapping in **Python**, resulting in reduction of distortion correction errors from 1.395 ± 0.136 pixels to 0.055 ± 0.192 pixels
- Developed mapping tables and leveraging **OpenCV** video processing algorithms in **C** to enhance DSP processing efficiency

PROJECTS

Rubik's Cube Helper November 2021 - November 2022

- Launched a dynamic web app with **React (JavaScript)** and created iOS client using **Swift** and **CocoaPods**
- Delivered **Node.js** backend with **Express**, hosted on **AWS EC2** for robust performance
- Designed and built **RESTful API** to enable CRUD and authentication, integrated with **MongoDB** for efficient data storage
- Implemented responsive design, accessibility development, and multiple interactions to meet WCAG 2.1 standard on web

EEG-based Brain-Computer Interface Research September 2019 – June 2021

- Constructed classification models of brain for material-specific Sternberg task based on 128-channel EEG dataset
- Extracted EEG features using empirical mode decomposition (EMD) and phase space reconstruction (PSR) in Matlab
- Employed machine learning such as SVM, KNN, Random Forest (RF) with nested cross-validation using Python and sklearn
- Improved classification performance from 71.45% to 91.82% and published results as first author in an IEEE paper

EDUCATION

Northeastern University Seattle, WA
Master of Science in Information Systems; GPA: 3.95/4.0 September 2021 – December 2023

Tongji University Shanghai, China
Bachelor of Engineering in Communications Engineering (Electrical Engineering) September 2017 – June 2021

ACADEMIC PUBLICATION

Classification of EEG signals during Working-Memory Maintenance based on Phase Space Reconstruction of Empirical Mode Decomposition, Oct 2020, ISBN:978-1-6654-2299-4